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Foldable display case

Abstract

A foldable display case includes: a case main body; first connection portions; second connection portions; first side-wall portions, each configured to be turnable between a first position where the first side-wall portion stands up from the case main body and a second position where the first side-wall portion falls down on the case main body, and each including first engagement portions; second side-wall portions, each configured to be turnable between a third position where the second side-wall portion stands up from the case main body and a fourth position where the second side-wall portion falls down on the case main body, and each including second engagement portions; and joint portions each including third engagement portions engageable with and disengageable from the first engagement portions of the turning first side-wall portion, and fourth engagement portions engageable with and disengageable from the second engagement portions of the turning second side-wall portion.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a foldable display case in which side walls are foldable.

BACKGROUND ART

(2) For example, a foldable display table also functioning as a storage box is known (for example, see Patent Literature 1). In the case of the display table, commodities are displayed while side walls in a longitudinal direction are folded, which makes it possible to display the commodities to be easily taken up by a customer visiting a store.

CITATION LIST

Patent Literature

(3) [Patent Literature 1] Japanese Patent Laid-Open No. 2013-31547

SUMMARY OF INVENTION

Technical Problem

(4) The display table is desirably used not only in a state where side walls in the longitudinal direction are folded but also in a state where side walls in a transverse direction are folded. However, in the case of the structure of the above-described existing display table, there is a problem that only the side walls in the transverse direction cannot be folded in terms of the structure.

(5) Accordingly, an object of the present invention is to provide a foldable display case in which a side wall in an optional direction is foldable.

Solution to Problem

(6) The above-described problems are solved by the present invention described below. A foldable display case according to the present invention (1) includes: a case main body having a square plate shape; first connection portions protruding from respective first sides along a transverse direction intersecting a longitudinal direction of the case main body, of an outer edge portion of the case main body; second connection portions protruding from respective second sides along the longitudinal direction, of the outer edge portion; first side-wall portions extending along the transverse direction, each configured to be turnable between a first position where the first side-wall portion stands up from the case main body and a second position where the first side-wall portion falls down on the case main body, around the corresponding first connection portion, and each including first engagement portions; second side-wall portions extending along the longitudinal direction, each configured to be turnable between a third position where the second side-wall portion stands up from the case main body and a fourth position where the second side-wall portion falls down on the case main body, around the corresponding second connection portion, and each including second engagement portions; and joint portions provided at respective corners of the case main body, and each including third engagement portions engageable with and disengageable from the first engagement portions of the turning first side-wall portion, and fourth engagement portions engageable with and disengageable from the second engagement portions of the turning second side-wall portion.

(7) A foldable display case according to the present invention (2) is the foldable display case according to (1) in which either one of the first side-wall portions and the joint portions includes a first latch and a first lever for moving the first latch, and the other of the first side-wall portions and the joint portions includes a first hole portion with which the first latch engages.

(8) A foldable display case according to the present invention (3) is the foldable display case according to (1) or (2) in which either one of the second side-wall portions and the joint portions includes a second latch and a second lever for moving the second latch, and the other of the second side-wall portions and the joint portions includes a second hole portion with which the second latch

engages when each of the second side-wall portions is at the third position.
(9) A foldable case according to the present invention (4) is the foldable case according to (1) in which each of the first connection portions includes fifth engagement portions, and the fifth engagement portions engage with the respective second side-wall portions positioned at the fourth position to hold the second side-wall portions at the fourth position.
(10) A foldable case according to the present invention (5) is the foldable case according to (1) or (2) in which the first connection portions protrude up to positions farther from the case main body than the second connection portions, and when each of the first side-wall portions is moved to the second position while the second side-wall portions are at the third position, a front end of each of the first side-wall portions is in contact with the case main body to form a slope going away from the case main body as it goes from the front end toward the corresponding first connection portion.
(11) A foldable case according to the present invention (6) is the foldable case according to (5) further including placement portions provided at respective corners of the case main body and on which the respective joint portions are placed, in which the placement portions protrude from the case main body up to a height equivalent to a height of the first connection portions, and the first connection portions inhibit the joint portions from falling down on the first connection portions together with the second side-wall portions.
(12) A foldable case according to the present invention (7) is the foldable case according to (1) or (2) in which the first connection portions protrude up to positions farther from the case main body than the second connection portions, the first side-wall portions are configured to fall down on the second side-wall portions positioned at the fourth position, and the joint portions are configured to fall down on the second connection portions together with the first side-wall portions.
(13) A foldable case according to the present invention (8) is the foldable case according to (7) in which each of the paired second side-wall portions includes a corner portion extending in the longitudinal direction, the corner portions are in contact with the respective second connection portions when the second side-wall portions are at the third position, and are separated from the case main body when the second side-wall portions are at the fourth position, each of the joint portions includes paired inner surfaces that form an L-shaped cross-section and are substantially orthogonal to each other, and the paired inner surfaces of the joint portions are in contact with the corner portions when the paired second side-wall portions are at the fourth position.
Advantageous Effect of Invention
(14) According to the present invention, it is possible to provide a foldable display case in which a side wall in an optional direction is foldable.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a perspective view illustrating a foldable display case according to an embodiment.
(2) FIG. **2** is a perspective view illustrating a state where joint portions are detached from the foldable display case illustrated in FIG. **1**.
(3) FIG. **3** is a perspective view illustrating a case main body of the foldable display case illustrated in FIG. **1** as viewed from a first surface side.
(4) FIG. **4** is a perspective view illustrating the case main body illustrated in FIG. **3** as viewed from a second surface side.
(5) FIG. **5** is a cross-sectional view taken along a position of line F**5**-F**5** illustrated in FIG. **1**.
(6) FIG. **6** is a perspective view illustrating a first side-wall portion of the foldable display case illustrated in FIG. **1** as viewed from outside.
(7) FIG. **7** is a perspective view illustrating the first side-wall portion illustrated in FIG. **6** as viewed from inside.

(8) FIG. **8** is a perspective view illustrating a second side-wall portion of the foldable display case illustrated in FIG. **1** as viewed from outside.
(9) FIG. **9** is a perspective view illustrating the second side-wall portion illustrated in FIG. **8** as viewed from inside.
(10) FIG. **10** is a perspective view illustrating four joint portions in total of the foldable display case illustrated in FIG. **1**.
(11) FIG. **11** is a cross-sectional view taken along a position of line F**11**-F**11** of the foldable display case illustrated in FIG. **1**.
(12) FIG. **12** is a cross-sectional view illustrating a state where, in the foldable display case illustrated in FIG. **11**, a second lever is pulled up in an arrow direction, to disengage a second latch from a second hole portion.
(13) FIG. **13** is a perspective view illustrating a state after one of paired second side-wall portions is moved to a fourth position in the foldable display case illustrated in FIG. **1**.
(14) FIG. **14** is a perspective view illustrating a state where one of paired first side-wall portions is moved to a second position in the foldable display case illustrated in FIG. **1**.
(15) FIG. **15** is a plan view of the foldable display case illustrated in FIG. **13** as viewed from above.
(16) FIG. **16** is a cross-sectional view taken along a position of line F**16**-F**16** of the foldable display case illustrated in FIG. **15**.
(17) FIG. **17** is a perspective view illustrating a folded state where the first side-wall portions and the second side-wall portions of the foldable display case illustrated in FIG. **1** fall down on the case main body.
(18) FIG. **18** is a perspective view illustrating foldable display cases in a plurality of use modes including a first special use mode **61**.
(19) FIG. **19** is a perspective view illustrating paired foldable display cases in a first applicable use mode.
(20) FIG. **20** is a perspective view illustrating paired foldable display cases in a second applicable use mode.
(21) FIG. **21** is a perspective view illustrating a first side-wall portion and a joint portion of a foldable display case according to a modification.
(22) FIG. **22** is a plan view illustrating a part of the foldable display case illustrated in FIG. **21**.
(23) FIG. **23** is a cross-sectional view taken along a position of line F**23**-F**23** of the foldable display case illustrated in FIG. **2**.
(24) FIG. **24** is a cross-sectional view illustrating a state where the first side-wall portion is moved to a first position in the foldable display case illustrated in FIG. **23**.

DESCRIPTION OF EMBODIMENT

(25) An embodiment of a foldable display case according to the present invention is described below with reference to drawings. The foldable display case according to the present invention is used not only to carry in commodities from a vehicle to a backyard of a store, but also to display the commodities by being installed in a sales floor of the store. Further, the foldable display case according to the present invention can be used to display the commodities while only side-wall portions in a transverse direction are folded. In addition, the foldable display case according to the present invention can be used to display the commodities while only side-wall portions in a longitudinal direction are folded, as in the existing display case.

Embodiment

(26) As illustrated in FIG. **1** to FIG. **4**, a foldable display case **11** includes: a case main body **12** having a square plate shape; paired first connection portions **14** protruding from respective first sides **13** along a transverse direction S, of an outer edge portion of the case main body **12**; paired second connection portions **15** protruding from respective second sides **19** along a longitudinal direction L, of the outer edge portion of the case main body **12**; paired first side-wall portions **16**

connected to the respective first connection portions **14**; paired second side-wall portions **17** connected to the respective second connection portions **15**; four joint portions **18** each interposed between the first side-wall portion **16** and the second side-wall portion **17**; fifth engagement portions **21** provided in the first connection portions **14**; and four placement portions **22** provided at respective corners of the case main body **12**. Each of the components of the foldable display case **11** is made of, for example, a synthetic resin material.

(27) As illustrated in FIG. **3** and FIG. **4**, the case main body **12** includes a first surface **12**A on a front side and a second surface **12**B on a rear side. The paired first connection portions **14** and the paired second connection portions **15** rise from the first surface **12**A. Therefore, the first connection portions **14** and the second connection portions **15** form a frame shape rising from the first surface **12**A. The first connection portions **14** are formed in the same form. The second connection portions **15** are also formed in the same form.

(28) As illustrated in FIG. **5**, each of the paired second connection portions **15** is formed in a groove shape into which second shaft portions **23** of the corresponding second side-wall portion **17** are fitted. Each of the second shaft portions **23** of the second side-wall portions **17** has a substantially columnar shape, and partially includes a protrusion **23**A. The protrusions **23**A engage with respective projections **24** of the second connection portions **15**. The second shaft portions **23** of the second side-wall portions **17** are not easily detached from the second connection portions **15** by the projections **24** and the protrusions **23**A.

(29) Each of the paired first connection portions **14** has a form similar to the form of each of the second connection portions **15**. In other words, each of the first connection portions **14** is formed in a groove shape into which first shaft portions of the corresponding first side-wall portion **16** are fitted. The transverse direction S in which the first connection portions **14** extend is a direction intersecting the longitudinal direction L in which the second connection portions **15** extend. Each of the first connection portions **14** protrudes up to a position farther from the case main body **12** than each of the second connection portions **15**.

(30) As illustrated in FIG. **3**, each of placement portions **22** is provided so as to rise from the first surface **12**A, and is formed so as to be continuous with and at the same protrusion height as the corresponding first connection portion **14**. The joint portions **18** are placed on the respective placement portions **22**. Each of the placement portions **22** has a rounded portion **22**A such that a corner portion on a center part side of the case main body **12** has an arc shape. The joint portions **18** placed on the respective placement portions **22** can easily fall down on the second connection portions **15** by virtue of the rounded portions **22**A.

(31) As illustrated in FIG. **8** and FIG. **9**, the paired second side-wall portions **17** are formed in the same form. The second side-wall portions **17** extend along the longitudinal direction L. Each second side-wall portion **17** includes: a second side-wall portion main body **25** having a flat plate shape; upper and lower paired second engagement portions **26** provided on each of side surfaces of the second side-wall portion main body **25**; second reinforcing plates **27** each having a flat plate shape and reinforcing the paired second engagement portions **26** so as to couple the paired second engagement portions **26**; second latches **28** each provided between the paired second engagement portions **26**; second levers **31** retracting the respective second latches **28**; and the second shaft portions **23** and a corner portion **32** provided on a lower end of the second side-wall portion main body **25**. Each second side-wall portion **17** is turnable between a third position P**3** where the second side-wall portion **17** stands up from the case main body **12** and a fourth position P**4** where the second side-wall portion **17** falls down on the case main body **12**, around the corresponding second connection portion **15** (second shaft portions **23**). The corner portion **32** extends in the longitudinal direction L. The corner portion **32** is in contact with the corresponding second connection portion **15** when the second side-wall portion **17** is at the third position P**1**, and is separated from the case main body **12** when the second side-wall portion **17** is at a second position P**2**.

(32) Each of the second engagement portions **26** is formed in a hook shape. Front ends of the

second engagement portions **26** extend in a thickness direction of the second side-wall portion **17**. The second engagement portions **26** are integrally formed with the corresponding second reinforcing plates **27**. As illustrated in FIG. **9**, FIG. **11**, and FIG. **12**, each of the second levers **31** includes a second finger hook portion **33**, and a second supporting portion **34** supporting the second latch **28** in a cantilever state. The second levers **31** are each integrally made of a synthetic resin material.

(33) The second levers **31** are housed in respective concave portions of the second side-wall portion main body **25** so as to be slidable to the second side-wall portion main body **25**. As illustrated in FIG. **11** and FIG. **12**, the second latches **28** are integrally formed with the respective second levers **31**, and are each formed in a trapezoidal shape. The second latches **28** can be retracted and protrude from respective opening portions **35**. As illustrated by an arrow in FIG. **12**, when a user pulls the second levers **31** upward through the respective second finger hook portions **33**, the second supporting portions **34** are each bent due to relationship between the opening portion **35** and the second latch **28** provided in the second side-wall portion main body **25**, thereby retracting the second latches **28** into the second side-wall portion **17**. In this state, the user can move the second side-wall portion **17** from the third position P**3** to the fourth position P**4**. In contrast, when the user releases the pulling up of the second levers **31**, the second supporting portions **34** are restored to original shapes before being bent. As a result, the second levers **31** are returned to original positions, and the second latches **28** protrude to the outside again. Note that the configuration of the second levers **31** is illustrative, and a configuration in which the second levers **31** are pushed down to retract the respective second latches **28** may be adopted.

(34) To return the second side-wall portion **17** from the fourth position P**4** to the third position P**3**, the second side-wall portion **17** is just moved to the third position P**3**. At this time, after the second latches **28** are retracted into the second side-wall portion **17** by action of inclined surfaces, the second supporting portions **34** are restored to the original shapes and the second latches **28** accordingly protrude again. As a result, the second latches **28** engage with respective second hole portions **56** of the joint portions **18**, and the second side-wall portion **17** is held at the third position P**3**.

(35) As illustrated in FIG. **6** and FIG. **7**, the paired first side-wall portions **16** are formed in the same form. The first side-wall portions **16** extend along the transverse direction S. The first side-wall portions **16** are different from the second side-wall portions **17** in that each of the first side-wall portions **16** has a length dimension different from a length dimension of each of the second side-wall portions **17**, and includes a handle portion **42**; however, the other configurations are substantially similar to the configurations of the second side-wall portions **17**.

(36) Each first side-wall portion **16** includes: a first side-wall portion main body **41** having a flat plate shape; the handle portion **42** provided at a lower center part of the first side-wall portion main body **41**; upper and lower paired first engagement portions **43** provided on each of side surfaces of the first side-wall portion main body **41**; first reinforcing plates **44** each reinforcing the paired first engagement portions **43** so as to couple the paired first engagement portions **43**; first latches **45** each provided between the paired first engagement portions **43**; first levers **46** retracting the respective first latches **45**; and first shaft portions **47** provided at a lower end of the first side-wall portion main body **41**. Each first side-wall portion **16** is turnable between a first position P**1** where the first side-wall portion **16** stands up from the case main body **12** and the second position P**2** where the first side-wall portion **16** falls down on the case main body **12**, around the corresponding first connection portion **14** (first shaft portions **47**).

(37) Each of the first engagement portions **43** is formed in a hook shape. Front ends of the first engagement portions **43** extend in a thickness direction of the first side-wall portion **16**. The first engagement portions **43** are integrally formed with the corresponding first reinforcing plates **44**. Each of the first levers **46** includes a first finger hook portion **48**, and a first supporting portion supporting the first latch **45** in a cantilever state. The first levers **46** are each integrally made of a

synthetic resin material.

(38) The first levers **46** are housed in respective concave portions of the first side-wall portion main body **41** so as to be slidable to the first side-wall portion main body **41**. The first latches **45** are integrally formed with the respective first levers **46**, and are each formed in a trapezoidal shape. The first latches **45** can be retracted and protrude from the respective opening portions **35**. When the user pulls the first levers **46** upward through the respective first finger hook portions **48**, the first supporting portions similar to the second supporting portions **34** are bent due to relationship between the opening portion **35** and the first latch **45**, thereby retracting the first latches **45** into the first side-wall portion **16**. In this state, the user can move the first side-wall portion **16** from the first position P**1** to the second position P**2**. In contrast, when the user releases the pulling up of the first levers **46**, the first supporting portions are restored to original shapes before being bent. As a result, the first latches **45** protrude to the outside again. Note that the configuration of the first levers **46** is illustrative, and a configuration in which the first levers **46** are pushed down to retract the second latches **28** may be adopted.

(39) To return the first side-wall portion **16** from the second position P**2** to the first position P**1**, the first side-wall portion **16** is just moved to the first position P**1**. At this time, after the first latches **45** are retracted into the first side-wall portion **16** by action of inclined surfaces, the first supporting portions are restored to the original shapes and the first latches **45** accordingly protrude again. As a result, the first latches **45** engage with respective first hole portions **55** of the joint portions **18**, and the first side-wall portion **16** is held at the first position P**1**.

(40) As illustrated in FIG. **10**, the four joint portions **18** are formed in the same form. The joint portions **18** are provided at the corners of the case main body **12**. Each of the joint portions **18** has an "L" shape as viewed from above, and is formed in a columnar shape extending in a direction intersecting the first surface **12**A of the case main body **12**. Each joint portion **18** includes: paired inner surfaces **50**; a first wall **51** facing the adjacent first side-wall portion **16**; a second wall **52** facing the adjacent second side-wall portion **17**; third engagement portions **53** engaging with the first engagement portions **43** of the adjacent first side-wall portion **16**; fourth engagement portions **54** engaging with the second engagement portions **26** of the adjacent second side-wall portion **17**; the first hole portion **55** provided in the first wall **51** and engaging with the first latch **45** of the adjacent first side-wall portion **16**; the second hole portion **56** provided in the second wall **52** and engaging with the second latch **28** of the adjacent second side-wall portion **17**; the positioning portion **57** provided at a top part: and a flat surface **58** provided on a bottom part.

(41) The third engagement portions **53** are formed in hole shapes into which the respective first engagement portions **43** each having a hook shape are inserted. The third engagement portions **53** can engage with and disengage from the first engagement portions **43** of the turning first side-wall portion **16**. The fourth engagement portions **54** are formed in hole shapes into which the respective second engagement portions **26** each having a hook shape are inserted. The fourth engagement portions **54** can engage with and disengage from the second engagement portions **26** of the turning second side-wall portion **17**. In the present embodiment, each of the third engagement portions **53** and the corresponding fourth engagement portion **54** are formed as an integrated cavity.

(42) The first hole portion **55** is formed as a through hole penetrating through the first wall **51**. The second hole portion **56** is formed as a through hole penetrating through the second wall **52**. Note that, in the present embodiment, the first latches **45** and the first levers **46** are provided in each of the first side-wall portions **16**, and the first hole portion **55** is provided in each of the joint portions **18**; however, the arrangement is not limited thereto. The first latch **45** and the first lever **46** may be provided in each of the joint portions **18**, and the first hole portions **55** may be provided in each of the first side-wall portions **16**.

(43) Likewise, in the present embodiment, the second latches **28** and the second levers **31** are provided in each of the second side-wall portions **17**, and the second hole portion **56** is provided in each of the joint portions **18**; however, the arrangement is not limited thereto. The second latch **28**

and the second lever **31** may be provided in each of the joint portions **18**, and the second hole portions **56** may be provided in each of the second side-wall portions **17**.
(44) Each of the positioning portions **57** includes a convex portion **57**A and a concave portion **57**B at adjacent positions. The positioning portions **57** constitute guides so as to enable the foldable display case **11** to be placed at a correct position when the foldable display case **11** in a normal use state where the first side-wall portions **16** and the second side-wall portions **17** are placed on the upper side is placed on the other foldable display case **11** in the normal use state where the first side-wall portions **16** and the second side-wall portions **17** are placed on the upper side.
(45) Likewise, the positioning portions **57** enable the foldable display case **11** in an upside-down state where the first side-wall portions **16** and the second side-wall portions **17** are placed on the lower side, to be installed on the foldable display case **11** in the normal use state where the first side-wall portions **16** and the second side-wall portions **17** are placed on the upper side, by virtue of shapes each obtained by combining the convex portion **57**A and the concave portion **57**B.
(46) As illustrated in FIG. **3**, the fifth engagement portions **21** are provided at positions close to the case main body **12** (first surface **12**A), in the first connection portions **14**. As illustrated in FIG. **3**, FIG. **15**, and FIG. **16**, each of the fifth engagement portions **21** is formed in a tongue shape protruding downward, and exerts flexibility at a front end side. Each of the fifth engagement portions **21** includes, at the front end part of the tongue shape, a locking projection **49** to lock the second reinforcing plate **27** of the corresponding side-wall portion **17**.
(47) Subsequently, action of the foldable display case **11** according to the present embodiment is described with reference to FIG. **1**, FIGS. **11** to **13**, and FIGS. **15** to **17**.
(48) As illustrated in FIG. **1**, in a state where the foldable display case **11** is set up as a box, the user can put commodities in the foldable display case **11** and transport the commodities, or can install the foldable display case **11** storing the commodities in a store to display the commodities. At this time, the user can carry the foldable display case **11** by using the handle portions **42**.
(49) As illustrated in FIG. **13**, commodities can be displayed while only one of the paired second side-wall portions **17** of the foldable display case **11** is moved from the third position P**3** to the fourth position P**4**. At this time, the user releases engagement by the second latches **28** by pulling up the second levers **31** of the second side-wall portion **17**, and moves the second side-wall portion **17** from the third position P**3** to the fourth position P**4**. At this time, engagement between the second engagement portions **26** of the second side-wall portion **17** and the fourth engagement portions **54** of the joint portions **18** is released. Further, when the second reinforcing plates **27** get over the respective locking projections **49** of the fifth engagement portions **21**, the second side-wall portion **17** is held at the fourth position P**4** by the fifth engagement portions **21**.
(50) Accordingly, for example, as in the first special use mode **61** illustrated in FIG. **18**, the foldable display case **11** in the upside-down state where the first side-wall portions **16** and the second side-wall portions **17** are placed on the lower side can be installed on the foldable display case **11** in the normal use state where the first side-wall portions **16** and the second side-wall portions **17** are placed on the upper side. Even in this case, in the foldable display case **11** in the upside-down state, a defect in which the second side-wall portion **17** returns from the fourth position P**4** to the third position P**3** by action of the gravity does not occur. This makes it possible to realize commodity display in which the front side is largely opened.
(51) To return the second side-wall portion **17** from the fourth position P**4** to the third position P**3**, the user raises the second side-wall portion **17** by a hand. As a result, the front ends of the fifth engagement portions **21** are bent and retracted, and the second reinforcing plates **27** can get over the respective locking projections **49** of the fifth engagement portions **21**. Each of the second latches **28** of the second side-wall portion **17** has an inclined surface so as to be gradually reduced in protrusion height as the position is changed from the fourth position P**4** to the third position P**3**. Therefore, when the user returns the second side-wall portion **17** from the fourth position P**4** to the third position P**3** without being conscious of presence of the second latches **28**, the second latches

28 are automatically retracted into the second side-wall portion **17**, and then return to the original protrusion states to engage with the second hole portions **56** of the joint portions **18**. As a result, the second side-wall portion **17** is held at the third position P**3**.

(52) As illustrated in FIG. **14**, commodities can be displayed while only one of the paired first side-wall portions **16** of the foldable display case **11** according to the present embodiment is moved from the first position P**1** to the second position P**2**. At this time, the user releases engagement by the first latches **45** by pulling up the first levers **46** of the first side-wall portion **16**, and moves the first side-wall portion **16** from the first position P**1** to the second position P**2**. At this time, since the first connection portion **14** protrudes up to the position farther from the case main body **12** than the second connection portions **15** as obvious from FIG. **14**, the first side-wall portion **16** at the second position P**2** forms a slope. In other words, the first side-wall portion **16** is in contact with the case main body **12** at a front end **16**A, to form the slope going away from the case main body **12** as it goes from the front end **16**A toward the first connection portion **14**.

(53) In such a use mode, for example, when elongated commodities such as Japanese radishes, burdocks, and green onions are displayed, the elongated commodities can be displayed while being partially placed on the slope.

(54) To return the first side-wall portion **16** from the second position P**2** to the first position P**1**, it is sufficient for the user to move the first side-wall portion **16** from the second position P**2** to the first position P**1**. Each of the first latches **45** of the first side-wall portion **16** has an inclined surface so as to be gradually reduced in protrusion height as the position is moved from the second position P**2** to the first position P**1**. Therefore, when the user returns the first side-wall portion **16** from the second position P**2** to the first position P**1** without being conscious of presence of the first latches **45**, the first latches **45** are automatically retracted into the first side-wall portion **16**, and then return to the original protrusion states to engage with the first hole portions **55** of the joint portions **18**. As a result, the first side-wall portion **16** is held at the first position P**1**.

(55) As illustrated in FIG. **17**, the foldable display case **11** can be put into a folded state. More specifically, by a method similar to the above-described method of moving the second side-wall portion **17** from the third position P**3** to the fourth position P**4**, each of the paired second side-wall portions **17** is moved from the third position P**3** where the second side-wall portion **17** stands up to the fourth position P**4** where the second side-wall portion **17** falls down on the case main body **12**. Further, each of the first side-wall portions **16** is moved from the first position P**1** to the second position P**2** so as to fall down on the second side-wall portions **17**. At this time, the first side-wall portions **16** and the joint portions **18** are smoothly moved from the first position P**1** to the second position P**2** by virtue of the rounded portions of the placement portions **22**. At this time, the paired inner surfaces **50** of the joint portions **18** are in contact with the corner portion **32** of the first side-wall portions **16** at the second position P**2**.

(56) On the other hand, a folded form of the foldable display case **11** in which the second side-wall portions **17** are positioned on the first side-wall portions **16** is prohibited. More specifically, as illustrated in FIG. **14**, in the state where the first side-wall portions **16** are at the second position P**2**, the second side-wall portions **17** do not fall down on the first side-wall portions **16** because the joint portions **18** serve as stoppers. As a result, occurrence of a defect in which the first side-wall portions **16** placed on the lower side are damaged by a load is prevented as much as possible.

(57) As illustrated in FIG. **19**, as a first applicable use mode of the foldable display case **11** according to the present embodiment, the foldable display cases **11** abut on each other while only one of the second side-wall portions **17** of each of the foldable display cases **11** is at the fourth position P**4**. This enables display of elongated commodities.

(58) Likewise, as illustrated in FIG. **20**, as a second applicable use mode of the foldable display case **11** according to the present embodiment, the foldable display cases **11** abut on each other while only one of the first side-wall portions **16** of each of the foldable display cases **11** is at the second position P**2**. This enables display of further elongated commodities.

(59) According to the present embodiment, the followings can be said. The foldable display case **11** includes: the case main body **12** having the square plate shape; the first connection portions **14** protruding from the respective first sides **13** along the transverse direction S intersecting the longitudinal direction L of the case main body **12**, of the outer end portion of the case main body **12**; the second connection portions **15** protruding from the respective second sides **19** along the longitudinal direction L, of the outer edge portion; the first side-wall portions **16** extending along the transverse direction S, each configured to be turnable between the first position P**1** where the first side-wall portion **16** stands up from the case main body **12** and the second position P**2** where the first side-wall portion **16** falls down on the case main body **12**, around the corresponding first connection portion **14**, and each including the first engagement portions **43**; the second side-wall portions **17** extending along the longitudinal direction L, each configured to be turnable between the third position P**3** where the second side-wall portion **17** stands up from the case main body **12** and the fourth position P**4** where the second side-wall portion **17** falls down on the case main body **12**, around the corresponding second connection portion **15**, and each including the second engagement portions **26**; and the joint portions **18** provided at the respective corners of the case main body **12**, and each including the third engagement portions **53** engageable with and disengageable from the first engagement portions **43** of the turning first side-wall portion **16**, and the fourth engagement portions **54** engageable with and disengageable from the second engagement portions **26** of the turning second side-wall portion **17**.
(60) According to the configuration, the second side-wall portions **17** along the transverse direction S are set at the fourth position P**4** while the second side-wall portions **17** along the longitudinal direction L are kept at the third position P**3**, which enables the commodities to be easily taken up. Further, the second side-wall portions **17** along the longitudinal direction L are set at the second position P**2** while the first side-wall portions **16** along the transverse direction S are kept at the first position P**1**, which also enables the commodities to be easily taken up. In particular, in a case of the former arrangement that is impossible so far, it is possible to realize display in which elongated commodities such as Japanese radishes, burdocks, and green onions are easily displayed and are easily taken up by a customer.
(61) In this case, either one of the first side-wall portions **16** and the joint portions **18** includes the first latch **45** and the first lever **46** for moving the first latch **45**, and the other of the first side-wall portions **16** and the joint portions **18** includes the first hole portion **55** with which the first latch **45** engages.
(62) According to the configuration, it is possible to prevent the first side-wall portions **16** from moving from the first position P**1** to the second position P**2** by action of the gravity at a time not intended by the user.
(63) In this case, either one of the second side-wall portions **17** and the joint portions **18** includes the second latch **28** and the second lever **31** for moving the second latch **28**, and the other of the second side-wall portions **17** and the joint portions **18** includes the second hole portion **56** with which the second latch **28** engages when each of the second side-wall portions **17** is at the third position P**3**.
(64) According to the configuration, it is possible to prevent the second side-wall portions **17** from moving from the third position P**3** to the fourth position P**4** by action of the gravity at a time not intended by the user.
(65) In this case, each of the first connection portions **14** includes the fifth engagement portions **21**, and the fifth engagement portions **21** engage with the respective second side-wall portions **17** positioned at the fourth position P**4** to hold the second side-wall portions **17** at the fourth position P**4**.
(66) According to the configuration, it is possible to hold the second side-wall portions **17** at the fourth position P**4**. Therefore, for example, even in a case where the foldable display case **11** is used in the upside-down state where the first side-wall portions **16** and the second side-wall

portions **17** are disposed on the lower side of the case main body **12**, it is possible to prevent the second side-wall portions **17** from moving from the fourth position P**4** to the third position P**3** by action of the gravity. Accordingly, the foldable display case **11** used in the upside-down state where the first side-wall portions **16** and the second side-wall portions **17** are disposed on the lower side of the case main body **12** is placed on the foldable display case **11** used in the normal use state where the first side-wall portions **16** and the second side-wall portions **17** are disposed on the upper side of the case main body **12**, which enables display in which the front side is largely opened.
(67) In this case, the first connection portions **14** protrude up to the positions farther from the case main body **12** than the second connection portions **15**. When each of the first side-wall portions **16** is moved to the second position P**2** while the second side-wall portions **17** are at the third position P**3**, the front end **16**A of each of the first side-wall portions **16** is in contact with the case main body **12** to form a slope going away from the case main body **12** as it goes from the front end **16**A toward the corresponding first connection portion **14**.
(68) According to the configuration, when elongated commodities such as Japanese radishes, burdocks, and green onions are displayed, the elongated commodities can be displayed while being partially placed on the slope. This enables not only three-dimensional display of the commodities, but also display in which the commodities are easily taken up by the customer. Further, two foldable display cases **11** each formed with the slope abut on each other to enable display of further elongated commodities.
(69) In this case, the foldable display case **11** includes the placement portions **22** provided at the respective corners of the case main body **12** and on which the respective joint portions **18** are placed. The placement portions **22** protrude from the case main body **12** up to a height equivalent to a height of the first connection portions **14**, and the first connection portions **14** inhibit the joint portions **18** from falling down on the first connection portions **14** together with the second side-wall portions **17**.
(70) According to the configuration, the second side-wall portions **17** do not fall down on the first side-wall portions **16** positioned at the second position P**2**. This makes it possible to reduce a risk causing a defect that the first side-wall portions **16** each forming the slope are damaged by loads from the second side-wall portions **17**.
(71) In this case, the first connection portions **14** protrude up to positions farther from the case main body **12** than the second connection portions **15**, the first side-wall portions **16** can fall down on the second side-wall portions **17** positioned at the fourth position P**4**, and the joint portions **18** can fall down on the second connection portions **15** together with the first side-wall portions **16**.
(72) According to the configuration, it is possible to compactly fold the foldable display case **11**.
(73) In this case, each of the paired second side-wall portions **17** includes the corner portion **32** extending in the longitudinal direction L, the corner portions **32** are in contact with the respective second connection portions **15** when the second side-wall portions **17** are at the third position P**1**, and are separated from the case main body **12** when the second side-wall portions **17** are at the second position P**2**, each of the joint portions **18** includes the paired inner surfaces **50** that form an L-shaped cross section and are substantially orthogonal to each other, and the paired inner surfaces **50** of the joint portions **18** are in contact with the corner portions **32** when the paired second side-wall portions **17** are at the fourth position P**4**.
(74) According to the configuration, the corner portions **32** can be disposed inside the inner surfaces of the joint portions **18**. This makes it possible to compactly fold the foldable display case **11**.
(75) Subsequently, the foldable display case **11** according to a modification is described with reference to FIG. **21** to FIG. **24**. In the following modification, differences from the above-described embodiment are mainly described, and description of portions common to the above-described embodiment is omitted.
(76) In the present modification, the first latches **45** and the second latches **28** are formed in the

same form. The first hole portions **55** and the second hole portions **56** are formed in the same form. Therefore, in the following, each of the first latches **45** and each of the first hole portions **55** illustrated in FIG. **21** and FIG. **23** are only described.
(77) Each of the first latches **45** is formed in, for example, a columnar shape, and is formed so as to protrude outward from the corresponding opening portion **35** of the first side-wall portion main body **41**. The first latches **45** can be vertically moved by operation of the respective first levers **46**. In the present modification, the first levers **46** are urged downward by respective unillustrated springs.
(78) Each of the first hole portions **55** is formed as a recess that has a hook-shaped cross-section and into which the corresponding second latch **28** can be fit.
(79) Subsequently, action of the foldable display case **11** according to the modification is described with reference to FIG. **21** to FIG. **24**. As illustrated in FIG. **21**, to return each of the first side-wall portions **16** from the second position P**2** to the first position P**1**, each of the first latches **45** is in contact with and runs on a corresponding slope **55**A provided near an entrance of the corresponding first hole portion **55**. As a result, each of the first latches **45** is housed in the corresponding first hole portion **55**. Further, at the same time, the first engagement portions **43** engages with the respective third engagement portions **53**. As a result, each of the first side-wall portions **16** is held at the first position P**1**.
(80) In contrast, to move each of the first side-wall portions **16** from the first position P**1** to the second position P**2**, the first lever **46** is pulled upward and the corresponding first latch **45** is accordingly moved upward. As a result, fitting of the first latch **45** into the corresponding first hole portion **55** is released. In this state, the user can move each of the first side-wall portions **16** from the first position P**1** to the second position P**2**.
(81) The above-described embodiment and modifications can be implemented while being added with various replacements and deformations. More specifically, in the above-described embodiment and modifications, the first connection portions **14** protrude to the positions farther from the case main body **12** than the second connection portions **15**; however, the form is not limited thereto. For example, the second connection portions **15** may protrude up to positions farther from the case main body **12** than the first connection portions **14** as a matter of course.
REFERENCE SIGNS LIST
(82) **11** foldable display case **12** case main body S transverse direction **13** first side **14** first connection portion L longitudinal direction **15** second connection portion **16** first side-wall portion **16**A front end **17** second side-wall portion **18** joint portion **19** second side **21** fifth engagement portion **22** placement portion **26** second engagement portion **28** second latch **31** second lever **32** corner portion P**3** third position P**4** fourth position **43** first engagement portion **45** first latch **46** first lever P**1** first position P**2** second position **50** inner surface **53** third engagement portion **54** fourth engagement portion **55** first hole portion **56** second hole portion

Claims

1. A foldable display case, comprising: a case main body having a square plate shape; first connection portions protruding from respective first sides along a transverse direction intersecting a longitudinal direction of the case main body, of an outer edge portion of the case main body; second connection portions protruding from respective second sides along the longitudinal direction, of the outer edge portion; first side-wall portions extending along the transverse direction, each configured to be turnable between a first position where the first side-wall portion stands up from the case main body and a second position where the first side-wall portion falls down on the case main body, around the corresponding first connection portion, and each including first engagement portions; second side-wall portions extending along the longitudinal direction, each configured to be turnable between a third position where the second side-wall portion stands

up from the case main body and a fourth position where the second side-wall portion falls down on the case main body, around the corresponding second connection portion, and each including second engagement portions; and joint portions provided at respective corners of the case main body, and each including third engagement portions engageable with and disengageable from the first engagement portions of the turning first side-wall portion, and fourth engagement portions engageable with and disengageable from the second engagement portions of the turning second side-wall portion, wherein each of the paired second side-wall portions includes a corner portion extending in the longitudinal direction, the corner portions are in contact with the respective second connection portions when the second side-wall portions are at the third position, and are separated from the case main body when the second side-wall portions are at the fourth position, each of the joint portions includes paired inner surfaces that form an L-shaped cross-section and are substantially orthogonal to each other, and the paired inner surfaces of the joint portions are in contact with the corner portions when the paired second side-wall portions are at the fourth position.

2. The foldable case according to claim 1, wherein either one of the first side-wall portions and the joint portions includes a first latch and a first lever for moving the first latch, and the other of the first side-wall portions and the joint portions includes a first hole portion with which the first latch engages.

3. The foldable case according to claim 2, wherein either one of the second side-wall portions and the joint portions includes a second latch and a second lever for moving the second latch, and the other of the second side-wall portions and the joint portions includes a second hole portion with which the second latch engages when each of the second side-wall portions is at the third position.

4. The foldable case according to claim 1, wherein the first connection portions protrude up to positions farther from the case main body than the second connection portions, and when each of the first side-wall portions is moved to the second position while the second side-wall portions are at the third position, a front end of each of the first side-wall portions is in contact with the case main body to form a slope going away from the case main body as it goes from the front end toward the corresponding first connection portion.

5. The foldable display case according to claim 4, further comprising placement portions provided at respective corners of the case main body and on which the respective joint portions are placed, wherein the placement portions protrude from the case main body up to a height equivalent to a height of the first connection portions, and the first connection portions inhibit the joint portions from falling down on the first connection portions together with the second side-wall portions.

6. The foldable display case according to claim 1, wherein the first connection portions protrude up to positions farther from the case main body than the second connection portions, the first side-wall portions are configured to fall down on the second side-wall portions positioned at the fourth position, and the joint portions are configured to fall down on the second connection portions together with the first side-wall portions.

7. A foldable display case, comprising: a case main body having a square plate shape; first connection portions protruding from respective first sides along a transverse direction intersecting a longitudinal direction of the case main body, of an outer edge portion of the case main body; second connection portions protruding from respective second sides along the longitudinal direction, of the outer edge portion; first side-wall portions extending along the transverse direction, each configured to be turnable between a first position where the first side-wall portion stands up from the case main body and a second position where the first side-wall portion falls down on the case main body, around the corresponding first connection portion, and each including first engagement portions; second side-wall portions extending along the longitudinal direction, each configured to be turnable between a third position where the second side-wall portion stands up from the case main body and a fourth position where the second side-wall portion falls down on the case main body, around the corresponding second connection portion, and each including

second engagement portions; and joint portions provided at respective corners of the case main body, and each including third engagement portions engageable with and disengageable from the first engagement portions of the turning first side-wall portion, and fourth engagement portions engageable with and disengageable from the second engagement portions of the turning second side-wall portion, wherein each of the first connection portions includes fifth engagement portions, and the fifth engagement portions engage with the respective second side-wall portions positioned at the fourth position to hold the second side-wall portions at the fourth position.